

Exercise 1:

1. Let $x, y \in \mathbb{R}$,

$$x * y = x + y + \sin(xy) = y + x + \sin(yx) = y * x$$

So the operation is commutative.

2. We search for e such that

$$x * e = x$$

$$\Leftrightarrow x + e + \sin(xe) = x$$

$$\Leftrightarrow \sin(xe) = -e.$$

$\Rightarrow e = 0$ as 0 is the solution that is true for all x in $\mathbb{R}$ and if such solution exist it is unique.

Taking $e = 0$ we have $\forall x \in \mathbb{R} \quad x * e = x$.

Then $*$ admit an identity element.

$$3. \quad f(-4) = -4 + 3 + \sin(-12) = -1 + \sin(-12) \\ \approx -1.20$$

$$f(-3.8) = -3.8 + 3 + \sin(3(-3.8)) = -0.8 + \sin(-3(-3.8)) \\ \approx -0.99$$

$$f(-3.1) \approx -1.16$$

So by the intermediate value theorem,

$$\exists \alpha \in]-4, -3.8[, \quad 3 * \alpha = f(\alpha) = -1$$

$$\exists \beta \in]-3.8, -3.1[, \quad 3 * \beta = f(\beta) = -1$$

1. We deduce that $3 * \alpha = 3 * \beta = -1$.

$$5. \quad (\alpha * 3) * \beta = -1 * \beta = \beta - 1 + \sin(-\beta) \\ = \beta - 1 - \sin(\beta)$$

$$\alpha * (3 * \beta) = \alpha * (-1) = \alpha - 1 - \sin(\alpha)$$

By assuming that the operation is associative, we obtain

$$\beta - 1 - \sin(\beta) = \alpha - 1 - \sin(\alpha)$$

$$\beta - \sin(\beta) = \alpha - \sin(\alpha)$$

However the function $f(x) = x - \sin(x)$ is strictly monoton

in $x \in]-4, -3[$ ($f'(x) = 1 - \cos(x)$) $\Rightarrow \alpha = \beta$ Contradiction

So $*$ is not associative.

Exercise 3:

1. (b)

i) The operation is binary operation.

By fixing $y \in]-1, 1[$ we consider the function

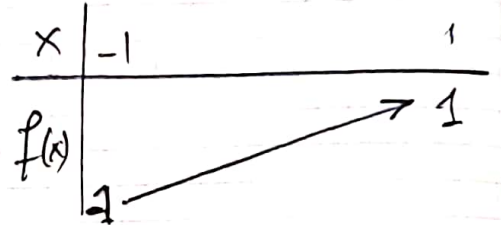
$$f_y(x) = \frac{x+y}{1+xy}$$

$$f'_y(x) = \frac{(1+xy) - y(x+y)}{(1+xy)^2} = \frac{1 + \cancel{xy} - \cancel{xy} - y^2}{(1+xy)^2} = \frac{1-y^2}{(1+xy)^2} > 0$$

So $f_y(x)$ is strictly increasing in $]-1, 1[$. Furthermore

$$f_y(-1) = -1 \quad f_y(1) = 1$$

we deduce $-1 < f_y(x) < 1$
 $-1 < x * y < 1$



Hence the operation is a binary operation.

ii) Associative: Let $x, y, z \in]-1, 1[$

$$\begin{aligned} (x * y) * z &= \frac{x+y}{1+xy} * z = \frac{\frac{x+y}{1+xy} + z}{1 + \frac{x+y}{1+xy} \cdot z} \\ &= \frac{\frac{x+y+z+xyz}{1+xy}}{1 + \frac{x+y+z+xyz}{1+xy}} = \frac{x+y+z+xyz}{1+xy+xz+yz} \end{aligned}$$

$$x * (y * z) = \frac{x+y+z+xyz}{1+xy+xz+yz} \quad (\text{You need to do computations that reach to this result})$$

iii) Identity element:

$$\left[\begin{array}{l} \text{We search for } e \in]-1, 1[\text{ s.t. } x * e = x \\ \frac{x+e}{1+xe} = x \Rightarrow x+e = x + x^2 e \\ \Rightarrow (1-x^2)e = 0 \Rightarrow e = 0 \end{array} \right]$$

Taking $e=0$ we have for all $x \in]-1, 1[$

$$x * e = x * 0 = \frac{x+0}{1+x \cdot 0} = x.$$

Note that the operation is commutative so no need to verify $x * y = y * x$.

(v) Commutative: Let $x, y \in]-1, 1[$.

$$x * y = \frac{x+y}{1+xy} = \frac{y+x}{1+yx} = y * x.$$

v) Inverse element:

$$\left[\begin{array}{l} \text{We search for } x^{-1} \text{ s.t.} \\ x * x^{-1} = 0 \Rightarrow \frac{x+x^{-1}}{1+xx^{-1}} = 0 \Rightarrow x+x^{-1} = 0 \\ x^{-1} = -x \end{array} \right]$$

Let $x \in]-1, 1[$, Choosing $x^{-1} = -x$ we

$$\text{have } x * x^{-1} = \frac{x+x^{-1}}{1+xx^{-1}} = 0.$$

Thus $(]-1, 1[, *)$ is an abelian group.

2. i) We know that $\forall A, B \in \mathcal{P}(S)$, $A \Delta B \subset S$ then the operation is a binary operation.

ii) Associative:

$$\text{Let } x \in (A \Delta B) \Delta C \Leftrightarrow x \in (A \Delta B) \cup C \text{ and } x \notin (A \Delta B) \cap C$$

$$\Rightarrow x \in (A \cup B) \cup C \text{ and } x \notin (A \cup B) \cap C$$

$$\Rightarrow x \in A \cup (B \cup C) \text{ and } x \notin (B \setminus A) \cap C$$

$$[(A \Delta B) \cap C = (A \setminus B \cup B \setminus A) \cap C = ((A \setminus B) \cap C) \cup ((B \setminus A) \cap C)]$$

$$\Rightarrow x \in A \cup (B \cup C) \text{ and } x \notin \overline{A \cup (B \cap C)}$$

$$[(B \setminus A) \cap C = B \cap \overline{A} \cap C = \overline{A \cup (B \cap C)}]$$

$$\Rightarrow x \in A \cup (B \cup C) \text{ and } x \in \overline{A \cup (B \cap C)}$$

$$\Rightarrow x \in A \cup ((B \cup C) \cap \overline{(B \cap C)})$$

$$\Rightarrow x \in A \cup (B \cup C \setminus B \cap C) \Rightarrow x \in A \cup (B \Delta C)$$

And if $x \in A \cap (B \Delta C) \Rightarrow x \in A$ and $x \in B \Delta C$

$$\Rightarrow x \in A \text{ and } x \in B \cup C \setminus B \cap C.$$

$$\Rightarrow \begin{cases} x \in A \text{ and } x \in B \Rightarrow x \in A \cap B \text{ - Contradiction} \\ x \in A \text{ and } x \in C \Rightarrow x \in A \cap C \text{ - Contradiction} \end{cases}$$

Hence $x \notin A \cap (B \Delta C)$

$$\Rightarrow x \in A \cup (B \Delta C) \cap \overline{(A \cap (B \Delta C))} \Rightarrow x \in A \Delta (B \Delta C)$$

$$\text{Thus } (A \Delta B) \Delta C \subset A \Delta (B \Delta C)$$

The other inclusion can be concluded similarly and we obtain

$$(A \Delta B) \Delta C = A \Delta (B \Delta C). \text{ Hence Associativity.}$$

iii) Commutative: Let $A, B \in \mathcal{P}(S)$

$$A \Delta B = A \cup B \setminus A \cap B = B \cup A \setminus B \cap A = B \Delta A.$$

iv) identity element:

Taking $\emptyset = \phi$ we have:

$$A \Delta \emptyset = A \cup \emptyset \setminus A \cap \emptyset = A \setminus \emptyset = A.$$

v) Inverse element: Let $A \in \mathcal{P}(S)$, choosing $A^{-1} = A$ we

$$\text{have } A \Delta A^{-1} = A \Delta A = A \cup A \setminus A \cap A = A \setminus A = \emptyset.$$

Therefore $(\mathcal{P}(S), \Delta)$ is an abelian group.

Exercise 8:

1. i) $(x, y) \oplus (x', y') = (x+x', y+y'+2xx') \in \mathbb{R}^2$ so the function is a binary operation.

$$\text{ii) } ((x, y) \oplus (x', y')) \oplus (x'', y'') = (x+x', y+y'+2xx') \oplus (x'', y'')$$

$$= ((x+x') + x'', y+y'+2xx' + y'' + 2(x+x')x'')$$

$$= (x+x'+x'', y+y'+2xx'+y''+2xx''+2x'x'')$$

$$(x, y) \oplus [(x', y') \oplus (x'', y'')] = (x+x'+x'', y+y'+y''+2xx'+2xx''+2x'x'')$$

(Computations left as an exercise)

Then associativity.

iii) Identity element:

$$\left[\begin{array}{l} \text{We search for } e \in \mathbb{R}^2 \text{ st } (x, y) \oplus (e_1, e_2) = (x, y) \quad \forall (x, y) \in \mathbb{R}^2 \\ (x+e_1, y+e_2+2xe_1) = (x, y) \Rightarrow \begin{cases} x+e_1 = x \Rightarrow e_1 = 0 \\ y+e_2 = y \Rightarrow e_2 = 0 \end{cases} \end{array} \right]$$

Taking $e = (0, 0)$, for all $(x, y) \in \mathbb{R}^2$

$$(x, y) \oplus (0, 0) = (x+0, y+0+2x \cdot 0) = (x, y).$$

iv) Inverse element:

$$\left[\begin{array}{l} \text{We search for } (x, y)^{-1} \text{ st } (x, y) \oplus (x, y)^{-1} = (0, 0) \\ (x+x', y+y'+2xx') = (0, 0) \Rightarrow \begin{cases} x+x' = 0 \Rightarrow x' = -x \\ y+y'+2xx' = 0 \Rightarrow \\ y+y'+2x(-x) = 0 \end{cases} \\ \Rightarrow y' = -y + 2x^2. \end{array} \right]$$

Let $(x, y) \in \mathbb{R}^2$, choosing $(x, y)^{-1} = (-x, -y + 2x^2)$ we obtain

$$\begin{aligned} (x, y) \oplus (x, y)^{-1} &= (x, y) \oplus (-x, -y + 2x^2) = \\ (x-x, y-y+2x^2+2x(-x)) &= (0, y-y+2x^2-2x^2) \\ &= (0, 0). \end{aligned}$$

I have forgotten to mention commutativity. This operation is also commutative and no need to compute $(x, y) \oplus (x, y)^{-1}$ and $(x, y)^{-1} \oplus (x, y)$. Finally $(\mathbb{R}^2, \oplus)$ forms an abelian group.

2.

i) $(0, 0) \in H$ for $x=0$.

ii) Let $u, v \in H$. Then $U = (x, x^2)$, $V = (y, y^2)$

$$\begin{aligned} (x, x^2) \oplus (y, y^2)^{-1} &= (x, x^2) \oplus (-y, -y^2 + 2y^2) \\ &= (x, x^2) \oplus (-y, y^2) \\ &= (x-y, x^2+y^2-2xy) \\ &= (x-y, (x-y)^2) \in H. \end{aligned}$$

So H is a subgroup of $(\mathbb{R}^2, \oplus)$.

3.

$$\begin{aligned} \Phi(x+y) &= (x+y, x^2+y^2+2xy) \\ &= (x, x^2) \oplus (y, y^2) = \Phi(x) \oplus \Phi(y). \end{aligned}$$

So Φ is a group homomorphism.

Injective: Let $x, y \in \mathbb{R}$, $\Phi(x) = \Phi(y) \Rightarrow (x, x^2) = (y, y^2)$
 $\Rightarrow \begin{cases} x=y \\ x^2=y^2 \end{cases} \Rightarrow x=y$.

Surjective: Let $U \in H$, choosing x st $U = (x, x^2)$ we have

$$\Phi(x) = (x, x^2) = U.$$

So Φ is a group isomorphism

$$\Phi^{-1}: H \longrightarrow \mathbb{R} \\ (x, x^2) \longrightarrow x$$

So Taking two elements $U, V \in H$ $U = (x, x^2), V = (y, y^2)$

$$\Phi(U \oplus V) = \Phi(U) \oplus \Phi(V)$$

$$\Rightarrow U \oplus V = \Phi(\Phi^{-1}(U) \oplus \Phi^{-1}(V))$$

$$\Rightarrow U \oplus V = \Phi(\Phi^{-1}(U) + \Phi^{-1}(V))$$

$$= \Phi(x+y) = (x+y, (x+y)^2).$$

Exercise 4: (a)

$$f_2 \circ f_1 = f_2, f_3 \circ f_1 = f_3, f_4 \circ f_1 = f_4, f_5 \circ f_1 = f_5, f_6 \circ f_1 = f_6$$

$$f_2 \circ f_2 = f_1, f_2 \circ f_3 = f_5, f_2 \circ f_4 = f_6, f_2 \circ f_5 = f_3$$

$$f_2 \circ f_6 = f_4.$$

$$f_3 \circ f_2 = f_4, f_3 \circ f_3 = f_1, f_3 \circ f_4 = f_2$$

$$f_3 \circ f_5 = f_6, f_3 \circ f_6 = f_5$$

$$f_4 \circ f_2 = f_3, f_4 \circ f_3 = f_6, f_4 \circ f_4 = f_5$$

$$f_4 \circ f_5 = f_1, f_4 \circ f_6 = f_2.$$

$$f_5 \circ f_2 = f_6, f_5 \circ f_3 = f_2$$

$$f_5 \circ f_4 = f_1, f_5 \circ f_5 = f_4, f_5 \circ f_6 = f_3$$

	f_1	f_2	f_3	f_4	f_5	f_6
f_1	1	2	3	4	5	6
f_2	2	1	5	6	3	4
f_3	3	4	1	2	6	5
f_4	4	3	6	5	1	2
f_5	5	6	2	1	4	3
f_6	6	5	4	3	2	1

(b) The composition of function is associative. f_1 is the identity element. Each element admit an inverse. The operation is binary operation (for every two elements of S their composition remain in S). Hence this table is a table of a group.

(b) Subgroups of S should have orders 1, 2, 3, or 6 by Lagrange's theorem.

Subgroups of order 1: $\langle f_1 \rangle = \{f_1\}$.

Subgroups of order 2: $\langle f_2 \rangle = \{f_2, f_1\}$, $\langle f_3 \rangle = \{f_3, f_1\}$,
 $\langle f_6 \rangle = \{f_6, f_1\}$.

Subgroups of order 3:

$$\langle f_4 \rangle = \{f_4, f_5, f_1\} = \langle f_5 \rangle = \langle f_4, f_5 \rangle.$$

$\langle f_2, f_3 \rangle = \{f_2, f_3, f_4, f_6, \dots\} = S = \langle f_2, f_4 \rangle = \langle f_2, f_6 \rangle = \langle f_2, f_6 \rangle$
 $= \langle f_3, f_4 \rangle = \langle f_3, f_6 \rangle = \langle f_4, f_6 \rangle = \langle f_4, f_6 \rangle$
 we obtain exactly 6 subgroups

2.

+	0	1
0	0	1
1	1	0

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

For order 2 this is the only possible Latin square then Cayley table for this order which is a group.

For order 3: Identity 0, each element has an inverse.

associativity

$$\begin{aligned}
 (1+2)+0 &= 0 = 1+(2+0) = (2+1)+0 = 2+(1+0) \\
 &= 0+(1+2) = (0+1)+2
 \end{aligned}$$

$$\begin{aligned}
 (1+2)+1 &= 0+1 = 1 = 1+(2+1) = (2+1)+1 = \\
 &2+(1+1) = (2+1)+1
 \end{aligned}$$

$$\begin{aligned}
 (2+1)+2 &= 2 = 2+(1+2) = (1+2)+2 = 1+(2+2) \\
 &= (1+2)+2
 \end{aligned}$$

⋮

Exercise 09:

$(G, \cdot)$ is group. H and K are subgroups of G .

1. $H \cap K$ is a subgroup

$$H \cap K = \{x \mid x \in K \wedge x \in H\}$$

1) $H \cap K \subset G$ since

$$H \cap K \subset H \subset G.$$

2) we know that

H is subgroup of G ,
so $e \in H$ also $e \in K$
since K is subgroup
of G ; therefore:
 $e \in H \cap K$.

3) the inverse:

- 1) $\forall x \in H$: the element x has
an inverse $x^{-1} \in H$ since
 H is subgroup of G .

2) $\forall y \in K$, the element y has
an inverse $y^{-1} \in K$, since
 K is a subgroup of G .

- if $x \in H \cap K \Rightarrow x \in K \wedge x \in H$
so $x^{-1} \in K \wedge x^{-1} \in H$
so $x^{-1} \in H \cap K$.

4. aHa^{-1} is a subgroup of G

$$\begin{cases} aHa^{-1} \subset G \\ aHa^{-1} \neq \emptyset \\ \forall x, y \in aHa^{-1} : xy^{-1} \in aHa^{-1} \end{cases}$$

• Let $x \in aHa^{-1}$. Then $x = ah_1a^{-1}$, $h_1 \in H$

Then $x \in G$ so $aHa^{-1} \subset G$

• For $e \in H$, we have $e = aea^{-1} \in aHa^{-1} \neq \emptyset$

• Let $x, y \in aHa^{-1}$. Then $\exists h_1, h_2 \in H$ such

$$\text{that } \begin{cases} x = ah_1a^{-1} \\ y = ah_2a^{-1} \text{ so } y^{-1} = ah_2^{-1}a^{-1} \end{cases}$$

$$\begin{aligned} \text{Then } xy^{-1} &= (ah_1a^{-1})(ah_2^{-1}a^{-1}) \\ &= a(h_1h_2^{-1})a^{-1} = ah_3a^{-1} \end{aligned}$$

So $xy^{-1} \in aHa^{-1}$. So aHa^{-1} is a subgroup of G

II - $\alpha \in \mathbb{R} \quad f_n : \mathbb{Z} \rightarrow \mathbb{C}^*$
 $p \mapsto e^{2\pi i n \alpha p}$

1. Let $p_1, p_2 \in \mathbb{Z}$

$$\begin{aligned} f_n(p_1 + p_2) &= e^{2\pi i n \alpha (p_1 + p_2)} \\ &= e^{2\pi i n \alpha p_1} \cdot e^{2\pi i n \alpha p_2} \\ &= f_n(p_1) \cdot f_n(p_2). \end{aligned}$$

Thus, $f_n \in \text{Hom}(\mathbb{Z}, \mathbb{C}^*)$

2. $\text{Im } f_n \subset U$

$$\begin{aligned} \text{Im } f_n &= \{ f_n(p) \mid p \in \mathbb{Z} \} \\ &= \{ e^{2\pi i n \alpha p} \mid p \in \mathbb{Z} \} \end{aligned}$$

Since $\forall p \in \mathbb{Z} \mid e^{2\pi i n \alpha p} \mid = 1$, then $\text{Im } f_n \subset U$

3. f_n injective iff $\alpha \notin \mathbb{Q}$

$\Rightarrow$) assume that f_n is injective. For the sake of a contradiction we assume that $\alpha \in \mathbb{Q}$

So $\alpha = \frac{a}{b}$ for some $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$

$$f_n(b) = e^{2\pi i n \frac{a}{b} b} = e^{2\pi i n a} = 1 \quad (n, a \in \mathbb{Z}).$$

So $0 \neq b \in \ker f_n$. Thus $\ker f_n \neq \{e_{\mathbb{Z}} = 0\}$

contradiction with the fact f_n injective

Therefore $\alpha \notin \mathbb{Q}$

$\Leftrightarrow$ assume that $\alpha \notin \mathbb{Q}$. we have

$$\ker f_n = \left\{ p \in \mathbb{Z} \mid f_n(p) = e_{\alpha} = 1 \right\}$$

$$= \left\{ p \in \mathbb{Z} \mid e^{2\pi i n \alpha p} = 1 \right\}$$

$$= \left\{ p \in \mathbb{Z} \mid 2\pi n \alpha p = 0 + 2k\pi \text{ for some } k \in \mathbb{Z} \right\}$$

$$k \in \mathbb{Z}$$

$$= \left\{ p \in \mathbb{Z} \mid n \alpha p = k \text{ for some } k \in \mathbb{Z} \right\}$$

$$= \{0\}$$

if $p \neq 0$, then, $\alpha = \frac{k}{n \cdot p} \in \mathbb{Q}$ which is impossible

Since $\ker f_n = \{0\}$, then f_n is injective

III - 2.

let $t_1, t_2 \in \mathbb{R}$ we have

$$\begin{aligned} \varphi(t_1 + t_2) &= e^{i(t_1 + t_2)} = e^{it_1} \cdot e^{it_2} \\ &= \varphi(t_1) \cdot \varphi(t_2). \end{aligned}$$

So $\varphi \in \text{Hom}(\mathbb{R}, \mathbb{T})$

on the other hand;

$$\ker f_n = \{ t \in \mathbb{R} \mid \varphi(t) = e_n = 1 \}$$

$$= \{ t \in \mathbb{R} \mid e^{it} = 1 \}$$

$$= \{ t \in \mathbb{R} \mid t = 0 + 2k\pi \text{ for some } k \in \mathbb{Z} \}$$

$$= \{ 2k\pi \mid k \in \mathbb{Z} \}$$

$$\ker \varphi_n \neq \{0\} = \{e_n\} \text{ so } \varphi \text{ is not injective}$$

$$\varphi(2\pi) = e^{i2\pi} = 1, \text{ so } 2\pi \in \ker \varphi.$$

So φ is not injective.

$$\text{Im } \varphi = \{ \varphi(t) \mid t \in \mathbb{R} \}$$

$$= \{ e^{it} \mid t \in \mathbb{R} \}$$

$$U = \{ z \in \mathbb{C} \mid |z| = 1 \}$$

$$= \{ z = 1 e^{i\theta} \mid \theta \in \mathbb{R} \}$$

φ is surjective.

$$\langle f_4 \rangle = \langle f_5 \rangle = \{ f_5^0 = f_1, f_5^1 = f_5, f_5^2 = f_4 \}$$

and S .

exercise 13:

$$U = \{ z \in \mathbb{C} \mid |z| = 1 \}, U_n = \{ z \in \mathbb{C} \mid z^n = 1 \}$$

$$U_\infty = \bigcup_{n \in \mathbb{N}^*} U_n.$$

1. $e_{\mathbb{C}} = 1$. since $|1| = 1$, then $1 \in U$ and $1 \in U_n$
 let $z_1, z_2 \in U$ so $|z_1| = 1$ and $|z_2| = 1$

$$\text{we have } |z_1 \cdot z_2^{-1}| = \frac{|z_1|}{|z_2^{-1}|} = 1. \text{ so}$$

$$z_1, z_2^{-1} \in U$$

in conclusion, $U \leq \mathbb{C}^*$.

$$2\mathbb{Z} \leq \mathbb{Z} \text{ and } 3\mathbb{Z} \leq \mathbb{Z}$$

$$\left. \begin{array}{l} 2 \in 2\mathbb{Z} \Rightarrow 2 \in 2\mathbb{Z} \cup 3\mathbb{Z} \\ 3 \in 3\mathbb{Z} \Rightarrow 3 \in 2\mathbb{Z} \cup 3\mathbb{Z} \end{array} \right\} 2+3 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$$

2. $e_{\mathbb{C}^*} = 1 \in U_1$ ($1^1 = 1$). so $1 \in U$ $U_n = \bigcup_{n \in \mathbb{N}^*} U_n$

let $z_1, z_2 \in U_\infty$. so $z_1 \in U_{n_1}$ for some $n_1 \in \mathbb{N}$ and $z_2 \in U_{n_2}$ for some $n_2 \in \mathbb{N}$, thus

$z_1^{n_1} = 1$ for some $n_1 \in \mathbb{N}$ and $z_2^{n_2} = 1$ for some $n_2 \in \mathbb{N}$
 we have $(z_1 \cdot z_2^{-1})^{n_1 n_2} = \frac{(z_1^{n_1})^{n_2}}{(z_2^{n_2})^{n_1}} = \frac{1^{n_2}}{1^{n_1}} = 1$

so $z_1 \cdot z_2^{-1} \in U_{n_1 n_2}$

so $z_1 \cdot z_2^{-1} \in \bigcup_{n \in \mathbb{N}} U_n$

3. $U_n \subset U_m$ iff n/m .

$\Leftarrow$) assume that n/m thus, $m = kn$ for some $k \in \mathbb{N}$.

let $z \in U_n$. so $z^n = 1$ we have:

$$z^m = z^{kn} = (z^n)^k = 1^k = 1$$

so $z \in U_m$. thus, $U_n \subset U_m$.

$\Rightarrow$) assume that $U_n \subset U_m$.

$$U_n = \left\{ e^{\frac{2i\pi k}{n}}, k = 0, \overline{n-1} \right\}$$

Since $U_n \subset U_m$, then.

$$\forall k = 0, \overline{n-1}, e^{\frac{2i\pi k m}{n}} = \left(e^{\frac{2i\pi k}{n}} \right)^m = 1$$

so

$$\forall k = \overline{0, n-1}, \quad \frac{2\pi k m}{n} = 0 + 2\pi k' \text{ for some}$$

$$k' \in \mathbb{Z}$$

Thus,

$$\forall k = \overline{0, n-1}, \quad \frac{k m}{n} = k', \quad k' \in \mathbb{Z}$$

$$\text{if } n \geq 2, \text{ then } \frac{m}{n} = k', \quad k' \in \mathbb{Z}$$

thus $n \mid m$.

$$\text{if } n = 1, U_1 = \{z \in \mathbb{C} \mid z^1 = 1\} = \{1\} \in U_m \text{ then}$$

and $1 \mid m$

So the property is true for $n=1$.

$$\begin{aligned} U_n &= \left\{ e^{\frac{2i\pi k}{n}} \mid k = \overline{0, n-1} \right\} \\ &= \left\{ \left(e^{\frac{2i\pi}{n}} \right)^k \mid k = \overline{0, n-1} \right\} \\ &= \left\langle e^{\frac{2i\pi}{n}} \right\rangle. \end{aligned}$$

Exo 14:

$$\forall x, y \in \mathbb{R}, x \star y = \sqrt[3]{x^3 + y^3}$$

$$x \star y = \sqrt[3]{x^3 + y^3} = \sqrt[3]{y^3 + x^3} = y \star x$$

$\star$ is commutative

$\star$ is associative

$e = 0$, sym of x is $-x$

so $(\mathbb{R}, \cdot)$ is abelian gp

we want to find $f: \mathbb{R} \rightarrow \mathbb{R}$, f is bijective

$$ST: f(xy) = f(x) + f(y)$$

we take $f: \mathbb{R} \rightarrow \mathbb{R}$
 $t \rightarrow t^3$

$$f \text{ satisfies } f(\sqrt[3]{x^3 + y^3}) = (\sqrt[3]{x^3 + y^3})^3 = x^3 + y^3$$

$$f(x) + f(y) = x^3 + y^3$$

we know $t \rightarrow t^3$ bijective

then $(\mathbb{R}, \cdot)$ is isomorphic to $(\mathbb{R}, +)$

Exercise 45:

I/1) Let $f: (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ homomorphisms:

$$\text{Then: } \begin{cases} f(0) = 0 \\ \forall n, m \in \mathbb{Z} : f(n+m) = f(n) + f(m) \end{cases}$$

Let $n \in \mathbb{N}$ Then

$$\begin{aligned} f(n) &= f((n-1)+1) = f(n-1) + f(1) \\ &= f((n-2)+1) + f(1) \\ &= f(n-2) + 2f(1) \\ &= \vdots \\ &= nf(1) \end{aligned}$$

We assume that $\exists n \in \mathbb{N} : f(n) = nf(1)$

$$\begin{aligned} \text{Then } f(n+1) &= f(n) + f(1) = nf(1) + f(1) \\ &= (n+1)f(1) \end{aligned}$$

• Let $n \in \mathbb{Z}^-$ Then:

$$\begin{aligned} f(0) &= f(n + (-n)) = f(n) + f(-n) \\ &= f(n) + (-n)f(1) \\ &= nf(1) - nf(1) = 0 \end{aligned}$$

So $\forall n \in \mathbb{Z} : f(n) = nf(1), f(1) \in \mathbb{Z}$

$$f(n) = kn$$

$\rightarrow$ If $k \neq 0$ Then $\forall n, m \in \mathbb{Z}$ such that:

$$\text{So } k(n-m) = 0$$

$$\text{Since } k \neq 0 \text{ Then } n = m$$

$$\bullet \text{ If } k = 0 \text{ Then } f(0) = f(1) = 0$$

$$\text{but } 0 \neq 1$$

$$\text{So } f \text{ is inj, if } k \neq 0$$

$\rightarrow$ Let $k \in \{-1, 1\}$ Then $\exists n \in \mathbb{Z}$,

$$f(n) = k+1, \text{ Then } kn = k+1$$

$$\text{a contradiction since } (k, k+1) = 1$$

So f is not surj

$$\bullet \text{ If } k = 1 \text{ Then } \forall m \in \mathbb{Z} : f(m) = m$$

$$\text{if } k = -1 \text{ Then } \forall m \in \mathbb{Z} : f(-m) = m$$

So f is surj, if $k \in \{1, -1\}$

2). Let $f: (\mathbb{R}, +) \rightarrow (\mathbb{Z}, +)$ homo

$$\text{Then } f \equiv 0$$

$$\begin{aligned} \frac{p}{q} &\longrightarrow f\left(\frac{p}{q}\right) = f\left(p \cdot \frac{1}{q}\right) \\ &= pf\left(\frac{1}{q}\right) \end{aligned}$$

$$\begin{cases} f(1) = n \\ f\left(\frac{n+1}{n+1}\right) = (n+1) \underbrace{f\left(\frac{1}{n+1}\right)}_K \end{cases}$$

$$\Rightarrow (n+1)K = n$$

II/ 1) Let $f: (\mathbb{R}, +) \rightarrow (\mathbb{R}^*, \times)$ isom

$$\text{Then: } \begin{cases} f \text{ homo} \\ f \text{ inj} \\ f \text{ surj} \end{cases}$$

$$\begin{aligned} \text{Let } x \in \mathbb{R} \text{ Then } f(x) &= f\left(\frac{x}{2} + \frac{x}{2}\right) \\ &= f\left(\frac{x}{2}\right) \times f\left(\frac{x}{2}\right) \\ &= f\left(\frac{x}{2}\right)^2 > 0 \end{aligned}$$

So it is not surj Then f is not isomorph

2) Let $f: (\mathbb{Q}, +) \rightarrow (\mathbb{Q}^*, \times)$ isom

Let $x \in \mathbb{Q}$, s.t. $f(x) = 2$ Then:

$$f\left(\frac{x}{2}\right)^2 \text{ so } f\left(\frac{x}{2}\right) = \sqrt{2} \in \mathbb{Q} \text{ (Contrad.)}$$

So f is not surj

Exercise 517:

*	a	b	c	d
a	a	a	a	a
b	a	b	c	d
c	a	a	a	a
d	a	b	c	d

$$b * c = ?$$

$$b * c = (c + d) * c = c * c + d * c$$

$$\text{les. } b * c = b * (d + b) = b * d + b * b =$$

$$- c * b = c * (c + d) = c * c + c * d$$

$$- c * b = (d + b) * b = d * b + b * b =$$

$$b + b = a$$

$$- c * c = (d + b) * c = d * c + b * c = c + ?$$

$$c * c = c * (d + b) = c * d + c * b$$

$$- d * d = d * (b + c) = d * b + d * c = b + c = d$$

$$b * c = b * (b + c) = b * b + b * c = b + c$$

$$+ b * c = b + c$$

Since $c * d = a \neq c = d * c$

Then R is not commutative

- R is a ring without

unity

we have $\mathbf{0}_R = a$.

$$a = \mathbf{0}_R.$$

1 time.

$$\left\{ \begin{array}{l} b \neq a = \mathbf{0}_R. \end{array} \right.$$

$$\left\{ \begin{array}{l} b + b = a = \mathbf{0}_R. \end{array} \right.$$

2 times.

$$\left\{ \begin{array}{l} c \neq a = \mathbf{0}_R. \\ c + c = a = \mathbf{0}_R. \end{array} \right\} \quad \left\{ \begin{array}{l} d \neq a = \mathbf{0}_R. \\ d + d = a = \mathbf{0}_R. \end{array} \right.$$

2 times

So $\forall r \in R, 2r = \mathbf{0}_R$ and

2 is the smallest positive

integer n such that $nr = \mathbf{0}_R$.

$$\text{so } \text{char}(R) = 2.$$

Exercise 19

I. 1. $(\mathbb{Z}/n\mathbb{Z}, \oplus, \odot)$ is a CR

We have $(\mathbb{Z}/n\mathbb{Z}, \oplus)$ is an AG and $\odot$ associative, binary, and we have:

$$\forall \bar{a}, \bar{b}, \bar{c} \in \mathbb{Z}/n\mathbb{Z} \quad \bar{a} \odot (\bar{b} \oplus \bar{c}) = (\bar{a} \odot \bar{b}) \oplus (\bar{a} \odot \bar{c})$$

and $\bar{a} \odot \bar{b} = \overline{ab} = \bar{b} \odot \bar{a}$ and $\forall \bar{c} \in \mathbb{Z}/n\mathbb{Z}$ then: $\mathbb{Z}/n\mathbb{Z}$ is a CR.

2. Note that: $\forall \bar{m} \in \mathbb{Z}/5\mathbb{Z} \quad 5 \cdot \bar{m} = \bar{0}$

$$\text{and } \begin{cases} 1\bar{1} = \bar{1} \neq \bar{0} \\ 2\bar{2} = \bar{4} \neq \bar{0} \\ 3\bar{2} = \bar{6} = \bar{1} \neq \bar{0} \\ 4\bar{1} = \bar{4} \neq \bar{0} \end{cases}$$

So $5 = \text{car}(\mathbb{Z}/5\mathbb{Z})$ similarly, $\text{car}(\mathbb{Z}/n\mathbb{Z}) = n$

3. $\mathbb{Z}/n\mathbb{Z}$ is integral domain $\Rightarrow n$ is prime

Suppose that $\mathbb{Z}/n\mathbb{Z}$ is an ID and $n = ab$:

$$1 < a, b < n$$

So $\bar{a} \bar{b} = \overline{ab} = \bar{n} = \bar{0}$ (Contradiction) since $\mathbb{Z}/n\mathbb{Z}$ is ID.

So $\mathbb{Z}/n\mathbb{Z}$ is ID $\Rightarrow n$ is prime

Now: n is prime $\Rightarrow \mathbb{Z}/n\mathbb{Z}$ is ID?

Suppose: $n = p$ is prime Let $\bar{a}, \bar{b} \in \mathbb{Z}/p\mathbb{Z}$

s.t. $\bar{a} \bar{b} = \bar{0}$ so $ab \in p\mathbb{Z}$

So $p \mid ab$ since p is prime, then

$$\begin{cases} p \mid a \\ p \mid b \end{cases} \text{ so } \begin{cases} a \in p\mathbb{Z} \\ b \in p\mathbb{Z} \end{cases} \text{ so } \begin{cases} \bar{a} = \bar{0} \\ \bar{b} = \bar{0} \end{cases}$$

So $\mathbb{Z}/p\mathbb{Z}$ is ID

II 1. 1) - we have: $0 = 0m + 0n \in m\mathbb{Z} + n\mathbb{Z} \neq \emptyset$

and $\forall \alpha = a_1m + b_1n, \beta = a_2m + b_2n \in m\mathbb{Z} + n\mathbb{Z}$

$$\alpha - \beta = (a_1 - a_2)m + (b_1 - b_2)n \in m\mathbb{Z} + n\mathbb{Z}$$

So $m\mathbb{Z} + n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$

$$1. \bar{m} = \{x \in \mathbb{Z} / m \mid x\} \quad \forall m \in n\mathbb{Z}$$

$$\bar{a} \cdot \bar{b} = \overline{ab} \quad , \quad \bar{0} = n\mathbb{Z} \quad , \quad \bar{1} = 1 + n\mathbb{Z}$$

$$(a + n\mathbb{Z})(b + n\mathbb{Z}) = ab + n\mathbb{Z}$$

$$(\bar{a} \odot \bar{b}) \odot \bar{c} = \bar{a} \odot (\bar{b} \odot \bar{c})$$

$$2. \text{Car}(A) = \min\{m \in \mathbb{N} / m\bar{n} = \bar{0}, \forall x \in A\}$$

$$m\bar{x} = m(x + n\mathbb{Z}) = mx + n\mathbb{Z}$$

$$= \bar{0} = \bar{m}x = \bar{0}, \forall x \in \mathbb{Z}/5\mathbb{Z}$$

2) we have to prove: $m\mathbb{Z} + n\mathbb{Z} = m\mathbb{Z} + (n+7m)\mathbb{Z}$

$$\text{Let } mk + (n+7m)k' \in m\mathbb{Z} + (n+7m)\mathbb{Z}$$

$$\text{So } m(k+7k') + nk' \in m\mathbb{Z} + n\mathbb{Z}$$

$$\text{Let } mk_1 + nk_2 \in m\mathbb{Z} + n\mathbb{Z} \text{ then let } r = k_1 - 7k_2$$

$$\text{So } m(7k_2 + r) + nk_2 = \quad \in \mathbb{Z}$$

$$mr + (n+7m)k_2 \in m\mathbb{Z} + (n+7m)\mathbb{Z}$$

$$\text{So } m\mathbb{Z} + n\mathbb{Z} = m\mathbb{Z} + (n+7m)\mathbb{Z}$$

III 1. 1) - we have:

$$x \in \frac{m}{n} \mathbb{Z} \iff x = \frac{m}{n} k, k \in \mathbb{Z}$$

$$\iff \begin{cases} x \in n\mathbb{Z} \\ x \in m\mathbb{Z} \end{cases} \iff x \in n\mathbb{Z} \cap m\mathbb{Z}$$

$$\text{So } m\mathbb{Z} = n\mathbb{Z} \cap m\mathbb{Z}$$

2) since $2, 3 \in 2\mathbb{Z} \cup 3\mathbb{Z}$ but $2+3=5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$

Then $2\mathbb{Z} \cup 3\mathbb{Z}$ is not a SG of $\mathbb{Z}$ then it's not a SR of $\mathbb{Z}$

3) we have $m\mathbb{Z} \cup n\mathbb{Z}$ is a SG of $\mathbb{Z}$

$$\text{iff: } \begin{cases} m\mathbb{Z} \subset n\mathbb{Z} \\ n\mathbb{Z} \subset m\mathbb{Z} \end{cases} \text{ iff } \begin{cases} n \mid m \\ m \mid n \end{cases}$$

4) Let G be a SG of $\mathbb{Z}$ (The smallest)

$$\text{s.t. } \begin{cases} m\mathbb{Z} \in G \\ n\mathbb{Z} \in G \end{cases}$$

Then by Bezout, $d = \gcd(n, m) = ma + nb \in G$, a.b.c

Then: $d\mathbb{Z} \subset G$

Since $\{m\mathbb{Z} \subset d\mathbb{Z}\}$ and $d\mathbb{Z} \subset G$

Ex 20

$$Z \oplus Z' = (a + a') + i(b + b')$$

$$Z \otimes Z' = aa' + i(ab' + ba')$$

We want to show that $(\mathbb{C}, \oplus, \otimes)$ is a commutative ring.

1. $(\mathbb{C}, \oplus)$ is an abelian group since $\oplus$ is nothing else than

The usual addition of $\mathbb{C}$

We have to show that $\forall Z, Z', Z'' \in \mathbb{C} \quad Z \otimes (Z' \otimes Z'') = (Z \otimes Z') \otimes Z''$
and $Z \otimes (Z' \oplus Z'') = Z \otimes Z' \oplus Z \otimes Z''$

$$(Z' \oplus Z'') \otimes Z = Z' \otimes Z \oplus Z'' \otimes Z$$

$$\text{it's clear that } Z \otimes Z' = aa' + i(ab' + ba')$$

$$Z' \otimes Z = (a' + b'i) \otimes (a + ib) = a'a + i(ab' + b'a)$$

Then it's commutative

We find the neutral element for $\otimes$

$Z_0 = (e + ie')$ is a neutral element if $\forall Z \in \mathbb{C}$

$$Z_0 \otimes Z = Z \otimes Z_0 = Z$$

$$(a + ib) \otimes (e + ie') = a + ib \Rightarrow ae = a \quad \forall a \quad e = 1$$

$$ae + i(ae' + eb)$$

$$ae' + be = b \Rightarrow e' = 0$$

$$\text{Then } Z_0 = 1$$

Thus $(\mathbb{C}; \oplus, \otimes)$ is a commutative ring.

2. Determining The set of invertible elements

$Z = a + ib$ is invertible if $Z \neq 0$

$$\exists Z' = a' + ib' \in \mathbb{C} \text{ st } Z \otimes Z' = 1$$

$$\Leftrightarrow Z \otimes Z' = (a + ib) \otimes (a' + ib') = aa' + (ab' + ba')i$$

$$= 1 + 0i \quad \text{if } a \neq 0 \text{ then } a' = \frac{1}{a}$$

$$\left\{ \begin{array}{l} aa' = 1 \quad (1) \\ ab' + ba' = 0 \quad (2) \end{array} \right.$$

$$\text{Then } b' = -\frac{b}{a}, \text{ so } Z' = \frac{1}{a} - i\frac{b}{a}$$

The set of invertible elements is

$$U = \{a+ib \mid a \neq 0\} = \mathbb{C} \setminus \{ib \mid b \in \mathbb{R}\} = \mathbb{C} \setminus i\mathbb{R}$$

3. The set of zero divisors

$Z = \{a+ib \in \mathbb{C} \mid a=0\}$ is a zero divisors if

$$\exists Z \neq 0 \text{ and } \exists Z' = a'+ib' \in \mathbb{C} \ (Z' \neq 0)$$

$$\begin{aligned} (a+ib) \otimes (a'+ib') &= 0 \\ &= aa' + i(ab' + ba') = 0 \end{aligned}$$

$$\begin{cases} aa' = 0 & (1) \\ ab' + ba' = 0 & (2) \end{cases}$$

$$aa' = 0 \Rightarrow a = 0 \ \forall a' = 0$$

$$\text{if } a = 0 \text{ Then } 0b' + ba' = 0$$

$$b \neq 0 \quad ba' = 0 \text{ but } b \neq 0 \text{ So } a' = 0$$

Conclusion. The set of zero divisors is $i\mathbb{R}$ (imaginary pure).

$$4. Z = a+ib \in \mathbb{C}$$

$$\forall n \in \mathbb{N}: Z^{(n)} = \underbrace{Z \otimes \dots \otimes Z}_{n \text{ times}}$$

a. Writing $Z^{(n)}$ as function of n : a and b .

$$Z^{(n)} = (a+ib)^n$$

$$Z^2 = (a+ib) \otimes (a+ib) = a^2 + (2ab)i$$

$$\begin{aligned} Z^3: Z^2 \otimes Z &= (a^2 + (2ab)i) \otimes (a+ib) \\ &= (a^3 + 3a^2bi) \end{aligned}$$

$$Z^4: Z^3 \otimes Z = (a^3 + 3a^2bi) \otimes (a+ib) = a^4 + 4a^3bi$$

We show by induction that

$$Z^n = a^n + na^{n-1}bi$$

→ we want to solve the equation.

$$Z^4 \oplus Z^2 = 6+1i$$

$$(a^4 + 3a^3bi) \oplus (a^2 + 2abi) = 6+1i$$

$$\begin{cases} a^4 + a^7 = 6 \\ 4a^3b + 2ab = 4 \end{cases}$$

Let $f = a^7$ Then $f^2 + f - 6 = 0$

$$f^2 + f - 6 = 0$$

Let E be a subring of R . Show that E is a subring

Show that 1 is following.

$a, b \in \mathbb{Z}$

$10^n \in G$,

condition,

Ex 23.1)

We have $1 = \frac{10}{10^1} \in R \neq \emptyset$, and

$\forall \frac{a}{10^n}, \frac{b}{10^m} \in R$, we have:

$$\frac{a}{10^n} - \frac{b}{10^m} = \frac{10^m a - 10^n b}{10^{n+m}} \in R \text{ and}$$

$$\frac{a}{10^n} \cdot \frac{b}{10^m} = \frac{ab}{10^{n+m}} \in R. \text{ So } R \text{ is a SR of } \mathbb{Q}.$$

2) Suppose that $\exists \frac{a}{10^n} = 1$, for some $\frac{a}{10^n} \in R$.

Then $3a = 10^n$. So $3/10$, a contradiction.

So 3 has no inverse in R . Conclusion: R is not a field.

3) Let $\frac{a}{10^n} \in U(R)$ be an invertible element in R .

Then $\exists \frac{b}{10^m} \in R$ s.t. $\frac{a}{10^n} \cdot \frac{b}{10^m} = 1$. So

Q. $ab = 10^{n+m}$. So $a/10^k$ $0 \leq k \leq n+m$. Then

$a = \pm 2^\alpha \cdot 5^\beta$, where $0 \leq \alpha, \beta \leq k$. So $U(R) = \left\{ \frac{\pm 2^\alpha \cdot 5^\beta}{10^n} \right\}$, $\alpha, \beta \in \mathbb{N}_0$

Exercise N° 26 (Boolean ring).

Let R be a ring, in which $\forall x \in R, x^2 = x$

1/ Example included examples:

$$(\mathbb{Z}_2, +, \cdot) \quad \mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z} = \{0, 1\}$$

$$0 = \{n \in \mathbb{Z} \mid n = 2n', n' \in \mathbb{Z}\}$$

$$1 = \{m \in \mathbb{Z} \mid m = 2m' + 1, m' \in \mathbb{Z}\}$$

$$0 \times 0 = 0 \quad / \quad 1 \times 1 = 1^2 = 1$$

* Let X be a nonempty set ($X \neq \emptyset$), $P(X)$: Power set of the set X

$$(P(X), \oplus, \otimes)$$

$$\forall A, B \in P(X)$$

$$A \oplus B = A \Delta B \quad , \quad A \otimes B = A \cap B$$

$$A \otimes A = A^2 \quad , \quad A \cap A = A$$

2/ We need to show that, $\forall a \in R, 2a = 0$

let a, b be two elements of R ($a \neq b$) we have $a+b \in R$,

$a \cdot b \in R$ (since R is a ring).

$$(a+b)^2 = (a+b)(a+b) = a^2 + ab + ba + b^2$$

$$\text{Since } a+b \in R \text{ so, } (a+b)^2 = a+b$$

$$\text{we get: } a+b = a+ab+ba+b$$

$$\text{for there, } ab+ba=0 \quad \text{--- ①}$$

$$\text{We put } b=1 \quad a \cdot 1 + 1 \cdot a = 0$$

$$a + a = 0$$

$$\text{--- } 2a = 0$$

3/ according to the relation note of ① we have: $\forall a, b \in R$,

$$ab+ba=0 \text{ i.e.}$$

$$ab = -ba \dots (2)$$

On other hand we have: $ba + ba = 2ba = 0$

$$\text{so, } ba = -ba \dots (3)$$

The conjunction of (2) and (3) gives: $ab = ba$

4/ We show that R cannot be reduced to three elements.

We assume that R have a three elements, i.e.,

$$R = \{0, 1, x\} \quad (x \neq 0, x \neq 1)$$

+	0	1	x
0	0	1	x
1	1	0	?
x	x	?	0

$$\begin{aligned} x+1 &= 0 \Rightarrow x=1 \quad \times \\ x+1 &= 1 \Rightarrow x=0 \quad \times \\ x+1 &= x \Rightarrow 1=0 \quad \times \end{aligned}$$

We can't build the addition table. Which is contradicts the assumption.
5/ Showing that R has Zero divisors let x, y be two element of R such that $x \neq y$ and $x \neq 0, y \neq 0$ let's consider

Let's consider $x \cdot y(x+y)$.

$$\begin{aligned} x \cdot y(x+y) &= x \cdot y \cdot x + x \cdot y \cdot y \\ &= x \cdot x \cdot y + x \cdot y \cdot y \\ &= x^2 \cdot y + x \cdot y^2 \\ &= x + y + xy \\ &= 2xy \\ &= 0 \end{aligned}$$

We have $x+y \neq 0$ since $x \neq y$ so, if $x \cdot y(x+y) = 0$ leads $x \cdot y = 0$ which implies that x and y are Zero divisors.

6/ the addition and multiplication in R is equivalent to the addition and multiplication in $\mathbb{Z}_2$. $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$

$$\begin{array}{c|c} 0^2=0 & 0+0=0 \\ 1^2=1 & 1+1=0 \end{array}$$

$$K \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \quad (|\mathbb{Z}_2 \times \mathbb{Z}_2| = 4).$$

$$R = \{0, 1, a, b\}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{0, 1\} \times \{0, 1\} = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$$

$0 = (0, 0)$ we define the addition and multiplication as follows

$$1 = (1, 1) \quad (x, y) + (z, t) = (x+z, y+t)$$

$$a = (1, 0) \quad (x, y) \cdot (z, t) = (xz, yt)$$

$$b = (0, 1)$$

$$0 + 0 = 0$$

$$1 + 1 = (1, 1) + (1, 1) = (1+1, 1+1) = (0, 0) = 0$$

$$1 + a = (1, 1) + (1, 0) = (0, 1) = b$$

$$1 + b = (1, 1) + (0, 1) = (1, 0) = a$$

$$a + a = (1, 0) + (1, 0) = (0, 0) = 0$$

$$\exists f: |R| = m, m \geq 2 \quad (m \in \mathbb{N})$$

We need to prove that $m = 2^n$

We can view R as a vector space over $\mathbb{Z}_2$ if $(e_1, e_2, \dots, e_m)$ for m a basis of R .

$$\text{So, } \forall x \in R, x = x_1 \cdot e_1 + x_2 \cdot e_2 + \dots + x_m \cdot e_m$$

$$x = (x_1, x_2, x_3, \dots, x_m).$$

$$\downarrow$$

$$2 \times 2 \times 2 \times \dots \times 2$$

We have 2^n elements in R .

Exercise 32:

1). Let $f: F \rightarrow F'$ a field homom

$$(\text{Ker } f = \{0\}) \bigcap_{i=1}^n \bigcap_{j=1}^n$$

Assume that f is not inj, Then:

$$\exists x, y \in F: x \neq y \wedge f(x) = f(y)$$

$$\begin{cases} x \neq y \Rightarrow x - y \neq 0 \\ \text{Let } z \in F \\ z = (x - y)(x - y)^{-1} z \Rightarrow f(z) = (f(x - y)(x - y)^{-1})z \\ f(z) = f(x - y) f((x - y)^{-1} z) \\ = (f(x) - f(y)) f((x - y)^{-1} z) \\ = 0 \end{cases}$$

$$\text{So: } f \equiv 0$$

So: any field homomorphism is either inj (monomorphism) or the null map

2. Let $x, y \in F \setminus \{0\}$, s.t. $xy = 0$

$$\text{Then } x^{-1}xy = x^{-1}0$$

So: $y = 0$ a contradiction.

So: F has no divisors of 0

So: F is an ID

3. Let $(0) \subsetneq I \subset F$ be an ideal.

Then $\exists x \in I \setminus \{0\}$

So: $x^{-1} \in F$ and $x^{-1}x = 1 \in I$.

Let $z \in F$ Then: $z \cdot 1 = z \in I$.

Then $F \subset I \subset F$ So: $F = I$.

4. Let R be a finite ID

Let $\alpha \in R \setminus \{0\}$ we have.

$$A = \{\alpha^n, n \in \mathbb{N}\} \subset R \text{ since } R \text{ is finite}$$

Then A is finite. Then $\exists n \in \mathbb{N}$, s.t.;

$$\alpha^n = \alpha. \text{ So } \alpha(\alpha^{n-1} - 1) = 0$$

$$\text{Since: } R \text{ is ID then } \begin{cases} \alpha = 0 \\ \alpha^{n-1} = 1 \end{cases}$$

$$\text{So: } \alpha^{n-1} = 1$$

$$\text{If } |A| = 1 \text{ Then } \alpha = 1. \text{ If } |A| \geq 2,$$

$$\text{So: } \exists y = \alpha^{n-2} \in R, \text{ s.t. } \alpha y = 1$$

Then α^{-1} exists

Then R is a field

$$\text{Let } \alpha \in R \setminus \{0\} \text{ Let } \alpha R = \{\alpha y / y \in R\} \subset R$$

Let $\phi: (R^*, +) \rightarrow (\alpha R, +)$ is homo of group

$$y \mapsto \alpha y$$

Let $y, z \in R$, s.t. $\phi(y) = \phi(z)$ Then $\alpha y = \alpha z$

Then $\alpha(y - z) = 0$ since R is ID Then $y = z$

So $|R| \leq |\alpha R|$ and $\alpha R \subset R$

$$|\alpha R| \leq |R|$$

So: $\alpha R = R$ Then $1 \in R = \alpha R$ Then $1 = \alpha y$ $y \in R$. So α^{-1} exists Then R is a field

exercise 33

$$R \subset K \text{ a subfield} \iff \begin{cases} \bigwedge x \in R \subset K \\ \forall x, y \in R: x-y \in R \\ \forall x, y \in R, y \neq 0: xy^{-1} \in R \end{cases}$$

1. we have $1 \neq 1+0\sqrt{3} \in \mathbb{Q}[\sqrt{3}] \subset \mathbb{R}$.

Let $a+b\sqrt{3}, a'+b'\sqrt{3} \in \mathbb{Q}[\sqrt{3}]$

2. suppose that: $a'+b'\sqrt{3} \neq 0$

$$a+b\sqrt{3}=0 \iff \begin{cases} b \neq 0 \implies \sqrt{3} = -\frac{a}{b} \in \mathbb{Q} \times \\ b=0 \text{ and } a=0 \end{cases}$$

$$\text{Then } a+b\sqrt{3} \neq 0 \implies \begin{cases} a \neq 0 \\ b \neq 0 \end{cases}$$

Then $a' \neq 0$ or $b' \neq 0$

$$\text{So } (a+b\sqrt{3}) \cdot (a'+b'\sqrt{3}) = \frac{a+b\sqrt{3}}{a'+b'\sqrt{3}} = d + e\sqrt{3} \in \mathbb{Q}[\sqrt{3}]$$

So: $\mathbb{Q}[\sqrt{3}]$ is a SF of $\mathbb{R}$

Infect: $\mathbb{Q}[\sqrt{n}]$ is a SF of $\mathbb{R}$.

$$2. a+b\sqrt{2}+c\sqrt{5}+d\sqrt{10}=0 \iff$$

$$a=b=c=d=0$$

$$(a+b\sqrt{2}) - (c\sqrt{5}+d\sqrt{10})$$

$$[(a+b\sqrt{2}) + (c\sqrt{5}+d\sqrt{10})][(a+b\sqrt{2}) - (c\sqrt{5}+d\sqrt{10})]$$

$$= a^2 + b^2\sqrt{2} + c^2\sqrt{5} + d^2\sqrt{10}$$

4. If $\{a+b\sqrt[3]{2} / a, b \in \mathbb{Q}\}$ is a SF of $\mathbb{R}$

$$\text{Then } \sqrt[3]{4} = (\sqrt[3]{2})^2 \in A \text{ so } \sqrt[3]{4} = a + b\sqrt[3]{2}$$

For some $a, b \in \mathbb{Q}$.

$$\text{So } 2\sqrt[3]{2} = a^2 + b^2\sqrt[3]{4} + 2ab\sqrt[3]{2} \dots (1)$$

if $b=0$ then $\sqrt[3]{4} = a \in \mathbb{Q}$

$$\text{So } 4 = \frac{a^3}{b^3}, a, b \in \mathbb{N}, (a, b) = 1, \text{ so:}$$

$$4b^3 = a^3 \text{ so } a = 2K, K \in \mathbb{Z}$$

$$\text{So } 4b^3 = 8K^3 \text{ so } b^3 = 2K^3 \text{ so } \frac{2}{b} \text{ (contra)}$$

$$\text{So } b \neq 0 \text{ so: } (1) \iff b^2\sqrt[3]{4} + 2(ab-1)\sqrt[3]{2} + a^2 = 0$$

$$(\text{So } \sqrt[3]{2} \text{ is a solution of } X^2 + \frac{2(ab-1)}{b^2}X + \frac{a^2}{b^2} = 0)$$

$$\text{So } b^2(a + \sqrt[3]{2}) + 2(ab-1)\sqrt[3]{2} + a^2 = 0$$

$$\text{Then } (b^2 + 2ab - 2)\sqrt[3]{2} + (ab^2 + a^2) = 0$$

$$\text{if } (b^2 + 2ab - 2) \neq 0$$

$$\text{Then: } \sqrt[3]{2} = \frac{-(ab^2 + a^2)}{b^2 + 2ab - 2} \in \mathbb{Q} \text{ (contradiction)}$$